

UltrON Industrial Monitoring Platform — Installation Guide

Overview

UltrON runs on a single Windows PC. All setup — stations, devices, parameters, SPCB push, RajAPI sync, license — is done through the web UI after launching the EXE. No separate server or Master PC needed.

Prerequisites

- Windows 10/11 (x64)
 - Admin rights (for LED board port 80 or RS485 serial, optional)
 - Internet connection (first-time license activation + RajAPI sync)
-

Part 1 — Install & Launch

1.1 Get UltrON.exe

Get the EXE from your admin or download from the release link. Place it anywhere (e.g. **C:\UltrON\UltrON.exe**).

1.2 Run

1. Double-click **UltrON.exe**
2. Terminal window opens — backend starts on <http://localhost:8000>
3. Browser opens automatically after a few seconds
4. Login as **Master** (password provided by admin)

1.3 Verify it works

- **Dashboard** — shows live telemetry once devices are configured
 - **Settings** → **Logs** — polling status, no errors
 - **Devices & Config** — device online, parameters reading
-

Part 2 — Configuration

Do these steps in the web UI after logging in as **Master**.

2.1 Create Station

1. Go to **Devices & Config** → **Stations** tab
2. Click **Add Station**

3. Enter:

- **Station Name** — your site name (e.g. "KTPP Plant")
- **Station Code** — short code for CPCB

4. Click **Save**

2.2 Add Device

1. Go to **Devices** tab → **Add Device**

2. Choose connection type:

- **Modbus TCP**: IP address, port (default 502), slave ID
- **Modbus RTU (serial)**: COM port, baud rate, slave ID
- **CSV file**: file path, delimiter, refresh interval

3. Click **Test Connection** to verify device is reachable

4. Click **Save**

2.3 Add Parameters

1. Select the device → **Add Parameter**

2. For each measurement:

- **Tag Name** — unique ID (e.g. **CO**, **S02**, **PM10**)
- **Register Address** — Modbus register or CSV column
- **Data Type** — **int16**, **float32**, **uint16**, etc.
- **Scale Factor / Offset** — raw value conversion
- **Unit** — **ppm**, **ppb**, **ug/m3**, **degC**, etc.
- **Min/Max Valid** — range for quality check
- **Alarm High/Low** — alert thresholds

3. Click **Test Read** to verify the parameter reads correctly

4. Click **Save**

2.4 Configure SPCB Push

1. Go to **Settings** → **CPCB** tab → **SPCB server** section

2. Click **Add SPCB Server**

3. Enter:

- **Name** — e.g. "GPCB Gujarat"
- **Live URL** — SPCB push endpoint
- **Delay URL** — SPCB delay endpoint (15 min retry)

4. Toggle server ON

5. In the mapping table below, map each parameter to a **CPCB Parameter** from the dropdown

6. Fill **API ID / Name / Password / vName** as provided by the PCB board

7. Click **Save**

- Click **Test Live / Test Delay** to confirm push works

2.5 Configure CPCB TXT Export

- In **Settings** → **CPCB** tab → **CPCB TXT File Generation** section
- Click **Add Config** for your station
- Set **Export Path** — folder for output files (e.g. **C:\UltrON\CPCB_Data**)
- Set **Retention Count** (default 97 files)
- Enable toggle
- Files auto-generate every 15 minutes in Annexure-I format

2.6 Configure LED Board (optional)

- In **Settings** → **CPCB** tab → **LED Board** tab
- Add LED server config, map parameters to channels
- Card URL: <http://<PC-LAN-IP>/api/v1/led?auth=username&PCB=1,2,3>
- Run EXE as **Administrator** if LED port 80 binding fails

2.7 Activate License

- Go to **Settings** → **License** tab
- Get AMC token from RajAPI admin
- Paste token → **Test & Activate**
- License activates **CENTRAL_API_KEY** — enables RajAPI sync

Part 3 — RajAPI Sync (Central Server)

UltrON communicates with **RajAPI** (rajapi.com) — the central admin panel to manage all client sites.

3.1 What syncs

Interval	Data	Direction
60s	Heartbeat (station name, device count, online/offline status, telemetry snapshot)	This PC → RajAPI
60s	Remote commands (restart polling, factory reset, config update)	RajAPI → This PC
60s (live)	SPCB telemetry push (latest 1-min values per mapped parameter)	This PC → PCB URL
900s (delay)	SPCB delayed push (backlog retry for failed live pushes)	This PC → PCB URL
15 min	CPCB TXT file generation (15-min averaged values, Annexure-I format)	This PC → Local file
On license verify	CENTRAL_API_KEY assignment (auth token for all RajAPI requests)	RajAPI → This PC

3.2 Auth flow

```
License activation → RajAPI returns CENTRAL_API_KEY
                        ↓
Heartbeat sends {gateway_id, device_secret, CENTRAL_API_KEY}
                        ↓
RajAPI validates → returns commands + sync acknowledgment
```

Auth methods checked in order:

1. **CENTRAL_API_KEY** — set by license activation (primary)
2. **GATEWAY_ID** + **DEVICE_SECRET** — from **.env** (secondary)
3. **RAJAPI_API_KEY** + **STATION_ID** — legacy (fallback)

3.3 Remote commands from RajAPI

RajAPI can send these commands via heartbeat response:

Command	Action
restart_polling	Restarts the Modbus/CSV polling engine
restart_app	Restarts the entire UltrON application
factory_reset	Clears all data and re-runs first-time setup
update_config	Pulls latest config from central server
reboot_pc	Reboots this PC

3.4 Monitoring on RajAPI

RajAPI dashboard shows for this PC:

- **Online/Offline** status (based on heartbeat recency)
- **Last seen** timestamp
- **Device count** and **parameter count**
- **AMC expiry** date
- **App version** (from UltrON EXE)
- **Recent SPCB push** status

3.5 Troubleshooting sync

Problem	Fix
Heartbeat 401	License expired or CENTRAL_API_KEY invalid. Reactivate license.
Heartbeat not reaching RajAPI	Check internet. RajAPI sync enabled in Settings.
Remote command not executing	This PC must be online. Command delivered on next 60s heartbeat.
SPCB push failing	Test from CPCB → Test Live. Check URL and PCB credentials.
CPCB TXT not generating	Verify Export Enabled + Export Path is writable.

Part 4 — Features Summary

Feature	Where	How
Live telemetry	Dashboard	WebSocket push every 5s
Polling	Background	5s cycle per device, dedup to 1-min storage
Trends	Trends screen	Historical chart with resolution selector
Reports	Reports screen	Normal (raw) + Average reports, PDF/CSV export
Alarms	Alarms screen	Configurable high/low thresholds, auto-trigger
CPCB TXT	Background	15-min averaged file in Annexure-I format
SPCB push	Background	JSON POST to PCB URL, live (60s) + delay (900s)
LED board	Port 80	HTTP GET with channel values
Calibration	Calibration screen	Multi-phase jobs with control charts
Broadcasts	Dashboard	Marquee + popup from RajAPI
RajAPI sync	Background	Heartbeat + commands every 60s
Live Trends	Dashboard modal	Real-time chart with 20-point sliding window
System logs	Settings → Logs	All engine events, push status, errors

Part 5 — Changing Configuration Later

When stations, devices, or mappings need to change:

1. Stop UltrON (close the window)
2. Get the latest **UltrON.exe** from your admin (contains updated config)
3. Replace the old EXE with the new one

- Restart UltrON — your data (telemetry, logs, alarms) is preserved in **ultron.db** next to the EXE
- Or if you have access to the admin panel on **rajapi.com**, Neeraj can push config changes remotely.
-

Appendix — Hardware Setup

RS485 / Modbus RTU (serial)

- Connect USB-to-RS485 converter to this PC
- Note COM port number from Device Manager
- In UltrON UI: Devices → select device → set port (e.g. **COM3**), baud rate, slave ID
- Restart polling or restart UltrON

Modbus TCP (Ethernet)

- Ensure this PC can reach the Modbus device over LAN
- In UltrON UI: Devices → select device → set IP, port (default 502), slave ID
- Restart polling or restart UltrON

LED Board

Run EXE as Administrator if LED server on port 80 is needed:

```
http://&lt;PC-LAN-IP&gt;/api/v1/led?auth=username&PCB=1, 2, 3
```

Appendix — Firewall / Network

UltrON listens on:

- **Port 8000** — main API + UI (browser access from LAN)
- **Port 80** — LED board endpoint (optional, admin rights required)
- **Modbus ports** — as configured per device (default 502 TCP)

Open Windows Firewall if LAN access to the UI is needed.

Appendix — Updating

- Stop UltrON (close window)
- Replace **ultron.exe** with new version
- Restart

The EXE auto-generates **secret.key** and **ultron.db** on first run if missing. On subsequent runs it uses the existing files — your data (historical telemetry, alarms, logs) persists across updates.

Appendix — Troubleshooting

Problem	Fix
EXE won't start	Check ultron_log.txt next to EXE. Common: port 8000 in use.
No telemetry data	Check device connection (serial/IP). Verify polling is active in Logs.
Login fails	Default: Master / password set during config. Reset by deleting ultron.db .
LED board not working	Run as Administrator.
RS485 not working	Run as Administrator. Check COM port in Device Manager.
SPCB push failing	Test push from CPCB screen. Check live/delay URLs.
AMC token expired	Contact Sunshine Technologies for renewal.
Config lost after update	Never delete ultron.db next to EXE — that file is your live data. Only the bundled DB inside the EXE is replaced.

Contact

Sunshine Technologies

- Support: 7659091468, 9133377852, 853
- Sales: 8801231166, 9133377852
- Web: <https://sunshinetechno.com/>